

Re-Analysis of Point 11 ("Susceptibility") with Precise Specification

TL;DR

- The precise definition is formally coherent as a discrete response function but is largely a re-packaging of (batch sharpness $\times$ gradient norm) divided by a feature-space density estimator; the numerator is essentially the Hessian-vector product norm $\eta \cdot \lambda_{\text{grad}}(\theta_t) \cdot \|g_t\|$, which has already been published as a loss-spike predictor (Bai et al., 2025) and is a close cousin of Andreyev & Beneventano's batch sharpness times the GraNd score.
 - The novel content is concentrated in three places: (i) the *coupling* of the curvature numerator to a model-defined feature-space entropy denominator, (ii) the *time-series Fourier/autocorrelation analysis* of S_t , and (iii) the *layer-indexed* variant; of these, (i) is genuinely under-explored, while (ii) overlaps non-trivially with the Notsawo et al. 2023 grokking spectral signature and the very recent Hennick & Corlouer 2026 spectral-heat-capacity work, and (iii) replicates much of what per-layer Fisher and per-layer sharpness already capture.
 - Verdict: pursue with substantial modifications. The strongest research question that survives cross-examination is not "is S_t a useful scalar?" but rather "does the *cross-spectrum* between a batch-sharpness-style numerator and a model-coupled feature-space entropy denominator carry early-warning information about phase transitions (grokking, EoS onset, plasticity loss) that batch sharpness alone, or representation-spectral-entropy alone, does not?"
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Key Findings

1. Numerator is well-understood and largely redundant. The first-order Taylor expansion $\Delta g_t = \nabla \ell_B(\theta_{t+1}) - \nabla \ell_B(\theta_t) \approx -\eta \nabla^2 \ell_B(\theta_t) g_t$ makes $\|\Delta g_t\| \approx \eta \cdot g_t^T H_B g_t / \|g_t\|$ when the gradient-direction Hessian is dominant — i.e., approximately $\eta \cdot \lambda_{\text{grad}}(\theta_t) \cdot \|g_t\|$, where λ_{grad} is the gradient-direction curvature. This exact quantity has been published as a loss-spike predictor (Bai et al., "Adaptive Preconditioners Trigger Loss Spikes in Adam," 2025 preprint), which proves that loss spikes occur *precisely* when λ_{grad} exceeds $2/\eta$. Andreyev & Beneventano's (ICLR/preprint 2024) "Batch Sharpness" is the same directional curvature but averaged over batches; they show it stabilises at $2/\eta$ under SGD. Hence $\|\Delta g_t\|$ is, up to a factor of η , batch-directional sharpness $\times$ gradient norm. (Zhiweibai) (arxiv)

2. Self-stabilization (Damian-Nichani-Lee, ICLR 2023) reframes Δg_t . Their cubic Taylor analysis shows that during EoS the sharpness self-stabilises through a third-order term that *predicts* gradient differences along the top Hessian eigenvector. So Δg_t is exactly the variable they implicitly track; its norm is large precisely during the catapult/self-stabilisation oscillations. This is *not* novel as a probe. (Princeton University) (Sitanchen)

3. Denominator $H_B(c)$ is the most distinctive piece, but its FORMAL coupling to the numerator has weak statistical-mechanics justification. A genuine susceptibility in stat-mech is $\partial(\text{observable})/\partial(\text{field})$, or via FDR, the ratio of fluctuation autocorrelation to response. Yaida (2018, ICLR 2019) and Han et al. (Phys. Rev. E 2021) derived FDRs for SGD that link weight-covariance to the gradient-Hessian product — these are the actual susceptibility analogues in ML, and they involve *weight*-statistics, not feature-space data entropy. Dividing a discrete response by a kernel-density estimate of feature-space marginal entropy is *not* the canonical susceptibility; it is a heuristic ratio. (arxiv + 2)

4. The Fourier component overlaps with published grokking work. Notsawo, Zhou, Pezeshki, Rish & Dumas (2023 preprint, "Predicting Grokking Long Before it Happens") explicitly compute the **Fourier spectral signature of training loss curves** and use the amplitude of low-frequency components to forecast grokking — this is the same methodology the proposal calls "novel." Hennick & Corlouer (2026 preprint, "From Density Matrices to Phase Transitions in Deep Learning") prove a *spectral heat capacity* derived from a 2-datapoint reduced density matrix gives early warning of second-order phase transitions in training **via critical slowing down**, again pre-empting the susceptibility-Fourier idea. Xu (2026 preprint, "Spectral Edge Dynamics") tracks rolling-window SVD of parameter updates as a phase-transition diagnostic. The proposal's autocorrelation/critical-slowing-down framing is therefore an active, fast-moving research front, not virgin territory — but the *specific* form $S_t = (\Delta g) / H(c)$ has not been Fourier-analysed. (arxiv + 5)

5. Layer-indexed $S_t(\ell)$ is largely covered by existing per-layer probes. Achille & Soatto's layer-wise Fisher trace (1711.08856), Lowell & Kastner's layer-wise Jacobian alignment (2024 preprint, NGA), Bolshim & Kugaevskikh's layer-wise Hessian spectra (2025 preprint), and Sankar et al.'s per-layer Hessian regulariser (2020) collectively occupy this niche. Per-layer $\|\Delta g_t(\ell)\|$ is essentially per-layer (sharpness $\times$ gradient-norm), which is computed in modern data-attribution pipelines (block-diagonal FIM in TRAK, GraSS, EK-FAC). (arxiv + 3)

6. The denominator-NTK approximation is weak. The proposal's claim that "feature-space marginal entropy $\approx$ NTK kernel-residual surprisal *if* penultimate features approximate the NTK top eigenspace" is only approximately true at convergence and after Neural Collapse / silent alignment. Atanasov, Bordelon & Pehlevan (ICLR 2022) show NTK eigenvectors evolve to align with target features only in the rich regime and primarily before substantial loss decrease (silent alignment). Lin (2025 preprint, "Feature Identification via the Empirical NTK") shows alignment between top eNTK eigenspaces and ground-truth features in toy models. Seleznova et al. (NeurIPS 2023, "Neural (Tangent Kernel) Collapse") prove the penultimate kernel and NTK share block structure *at convergence*, not generically during training. So the substitution $H_B(c) \leftrightarrow$ kernel-residual surprisal is provably tight only in restricted regimes. (arXiv) (arxiv)

Details

(1) Correctness as a susceptibility

Coherent as a *response function*, not strictly as a stat-mech susceptibility. Statistical-mechanics susceptibilities (Kubo, Onsager) are ratios of response to external perturbation, related to fluctuation autocorrelations via FDR. Yaida (1810.00004) and Han, Park, Lee & Han (Phys. Rev. E 104, 034126, 2021) give explicit FDRs for SGD: in the stationary regime, weight covariance $\leftrightarrow$ Hessian $\times$ noise covariance. The proposal's S_t is *not* this; it is the discrete response of the gradient to its own update — a self-response — divided by a coupled data-side entropy. This is a **discrete dimensional analogue** of a generalised susceptibility (response/entropy), but it does not satisfy any FDR identity and is not derived from a Hamiltonian-like energy.

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Discrete-time HVP interpretation is correct and tight. Under one update step $\theta_{t+1} = \theta_t - \eta g_t$, second-order Taylor gives $\Delta g_t = -\eta H_B(\theta_t) g_t + O(\eta^2 \|g_t\|^2)$. So:

- $\|\Delta g_t\| \approx \eta \cdot \sqrt{g_t^T H_B^2 g_t} \approx \eta \cdot \lambda_{\text{grad}}(\theta_t) \cdot \|g_t\|$ when the top Hessian eigenvector dominates the gradient direction (true at EoS — Cohen et al. 2021; Damian-Nichani-Lee 2023).
- This makes $S_t \approx \eta \cdot (\text{batch-directional sharpness}) \cdot \|g_t\| / H_B^{(c)}(\theta_t)$.

The numerator is therefore *exactly* the Bai et al. 2025 loss-spike predictor $\lambda_{\text{grad}} \cdot \|g_t\|$, scaled by η . Andreyev & Beneventano define batch sharpness as the directional curvature of mini-batch Hessians along stochastic gradients — i.e., λ_{grad} averaged over batches — and prove it saturates at $2/\eta$ under SGD ("Edge of Stochastic Stability"). For Adam, Bai et al. 2025 show this same quantity exceeds $2/\eta$ to trigger loss spikes. [arxiv + 2](#)

(2) Relevance & prior art

Quantity	Status vs. S_t
GraNd / EL2N (Paul et al. 2021)	$\ g_t\ $ alone, per-example. Subsumed by FLAG numerator.
Batch sharpness (Andreyev-Beneventano 2024)	Mini-batch directional curvature. Numerator of $S_t = \eta \cdot \text{batch-sharpness} \cdot \ g\ $.
λ_{grad} (Bai et al., 2506.04805, 2025)	Gradient-direction Hessian eigenvalue; predicts Adam loss spikes. Direct match for $\ \Delta g_t\ /\eta/\ g\$.
ZClip (Kumar et al., 2504.02507, 2024) / GradientStabilizer (2502.17055, 2026)	Adaptive thresholding on running gradient-norm statistics. (arxiv) (arXiv) Operates on $\ g_t\ $, not $\ \Delta g_t\ $.
Yaida FDR (1810.00004)	Weight-covariance / Hessian relations at stationarity. Different ratio.
Lindner FDRs (2204.00549) — neuroscience	Spike-train susceptibility; uses spontaneous power spectrum in denominator. (PubMed Central) Closer in spirit to S_t's spectrum analysis than ML literature.
Influence functions / TRAK / EK-FAC	Per-example $(\text{Hessian}^{-1} \cdot g)$ — <i>inverse</i> curvature $\times$ gradient, not curvature $\times$ gradient. Different niche.
Self-stabilization cubic term (Damian-Nichani-Lee ICLR 2023)	Δg_t along top eigenvector is exactly the negative-feedback variable. Numerator is implicit in their analysis.
2RDM spectral heat capacity (Hennick-Corlourer 2603.29805, 2026 preprint)	Coupled feature-side observable; tracks critical slowing down. Different but comparably-motivated quantity.
Spectral entropy of penultimate covariance (Truong et al., 2604.13123, 2026 preprint)	Order parameter for grokking. Closely related to S_t's denominator.
Notsawo spectral signature (2306.13253, 2023 preprint)	Fourier amplitude of loss curve to predict grokking. Direct match for the Fourier analysis of S_t.

The composite **(curvature $\times$ gradient norm) / (feature-space entropy)** with feature space defined by current penultimate representations does not appear, to my search, as a single named quantity in the published 2023–2026 training-dynamics literature. Components are separately well-studied; their *quotient* is not.

(3) Predictive value

The decomposition $S_t \approx \eta \cdot \lambda_{\text{grad}}(\theta_t) \cdot \|g_t\| / H_{B^{\wedge}(c)}(\theta_t)$ directly addresses the question:

- **Numerator vs. existing scores.** The product (batch sharpness)(gradient norm) is *almost exactly* the conjunctive predictor that several recent papers have converged on. Bai et al. 2025 show $\lambda_{\text{grad}} \cdot \|g\|$ alone forecasts Adam loss spikes; ZClip and GradientStabilizer operate on $\|g_t\|$ statistics; AdaGC blends gradient-norm history with curvature proxies. So the numerator alone offers little marginal information over $\|g_t\|^2 \cdot \cos^2(g_t, \text{top-eigenvector})$ — a quantity already monitored. (arXiv)
- **Denominator's role.** Dividing by feature-space marginal entropy could *only* add information if the denominator decouples from the numerator in interesting regimes — e.g., during silent alignment (numerator small, kernel structure changing), grokking late stages (numerator small but coverage entropy collapsing per Truong et al. 2026), or plasticity loss (numerator stable, but feature-space coverage dropping per Dohare et al. Nature 2024).
- **Penultimate-features $\approx$ NTK-eigenspace approximation.** Holds tightly (i) at convergence after Neural Collapse, (ii) in the silent-alignment phase (Atanasov-Bordelon-Pehlevan 2022) where eigenstructure changes while loss is flat, and (iii) in narrow toy settings (Lin 2025). It does **not** hold uniformly during the lazy-to-rich transition (Kumar et al. ICLR 2024); during catapult phases (Lewkowycz et al. 2020); or for finite-width feature-learning networks during early training (Atanasov et al., "Onset of Variance-Limited Behavior," ICLR 2023). Therefore identifying $H_B(c)$ with σ_{NTK} is at best a *late-training* approximation; the two will diverge in precisely the phases (early training, grokking onset, plasticity loss) where one would want the susceptibility to deliver early-warning information.
- **Verdict on novelty.** The composite is **mostly redundant in its scalar form** — replaceable by (batch sharpness)(gradient norm) for stability monitoring, and by representation-space spectral entropy for phase-transition forecasting. It is **plausibly non-redundant** when its *time-series statistics* — not its instantaneous value — are computed and combined: the joint power spectrum of numerator and denominator could capture phase-locking between curvature dynamics and feature-coverage dynamics, which no single scalar tracks today.

(4) Fourier / correlation analysis of training-dynamics time series

This is the most active overlap with very recent (2023–2026) literature:

- **Notsawo et al., "Predicting Grokking Long Before it Happens"** (2306.13253, 2023, preprint, has been heavily cited): explicitly computes the **Fourier transform of the training-loss curve** and uses low-frequency amplitude as a forecasting score for grokking. This is the closest published methodological match to the proposal's Fourier-of-S_t plan. Note: preprint, not formally peer-reviewed but well-cited. (arXiv)
- **Hennick & Corlouer 2026** (2603.29805, preprint, March 2026): the spectral heat capacity of the 2RDM is **proven** to provide early warning of second-order phase transitions in training via critical slowing down — explicitly importing the variance-and-lag-1-autocorrelation framework from Scheffer-style ecology/physics tipping-point literature. (arxiv) (Nature)
- **Xu, "Spectral Edge Thesis" and "Spectral Edge Dynamics of Training Trajectories"** (2603.28964, 2603.15678, 2026 preprints): rolling-window SVD spectral gap dynamics precede grokking events 24/24 with weight decay; signal/noise boundary in parameter-update spectra. (arxiv + 2)
- **Hertz & Tyrcha** (2412.10094, 2024 preprint, J. Stat. Mech.-style): explicitly identifies **critical slowing down** in deep recurrent networks near the interpolation transition, with learning times scaling like 1/(w-w_c). Provides direct theoretical grounding for autocorrelation-based phase-transition diagnostics. (arxiv)
- **Truong et al. "Spectral Entropy Collapse"** (2604.13123, 2026 preprint): normalised spectral entropy of representation covariance is a scalar order parameter crossing a threshold ~1,000 steps before grokking generalisation in 100% of runs. (arXiv)
- **Catapult oscillations and EoS oscillations** (Lewkowycz-Bahri-Dyer-Sohl-Dickstein-Gur-Ari 2020, Cohen et al. ICLR 2021, Damian-Nichani-Lee ICLR 2023) provide quasi-periodic structure that *would* show up in Fourier spectra of sharpness and gradient norm.
- **Yaida 2018 FDRs** are time-correlation identities by construction and have been used to *adaptively schedule* learning rates from observed weight-correlation statistics.

The Fourier analysis of {S_t} as defined — coupled time series of curvature-times-gradient-norm divided by feature-coverage entropy — has not been published. But the *family* of methods (Fourier / autocorrelation / spectral diagnostics on training time series) has roughly 5–10 closely related preprints in 2023–2026. The proposal would be entering a crowded niche.

(5) Layer-indexed S_t^(ℓ)

Per-layer analogues exist for every component of S_t individually:

- **Per-layer Fisher trace** (Achille & Soatto 1711.08856; ICLR 2019): canonical per-layer "effective connectivity" used for critical-period analysis.
- **Per-layer Hessian spectra** (Sankar et al. 2012.03801, AAAI 2021; Bolshim & Kugaevskikh 2510.17486, 2025 preprint): full per-layer eigenspectrum during training.
- **Layer-wise Jacobian alignment** (Lowell & Kastner 2406.00127, 2024 preprint): shows EoS is *caused* by per-layer Jacobian rotation, with peak alignment scaling with dataset size as a power law ($R^2=0.74-0.98$). (arxiv)
- **Layer-wise NTK / eNTK localisation** (Lin 2510.00468, 2025 preprint): layerwise eNTK localises features and diagnoses grokking phase transition. (arxiv)
- **Per-layer gradient noise scale** (Hilton et al., 2411.00999): GNS strongly correlated across layer types in transformers. (arxiv)
- **Block-diagonal FIM in attribution** (TRAK, GraSS, EK-FAC, LoGRA): treats per-layer curvature as the central object.

Per-layer $\|\Delta \mathbf{g}_t^{(\ell)}\| \approx \eta \cdot (\text{per-layer directional curvature}) \cdot (\text{per-layer gradient norm})$ is therefore a recombination of three already-monitored per-layer quantities. The information it adds beyond per-layer Fisher and per-layer sharpness is: it captures *interaction* between curvature and gradient direction *at the layer level*. This is non-trivial information but is partially captured by per-layer GNS and by per-layer Hessian-vector products in second-order optimisers (KFAC, Shampoo). The marginal value is small.

(6) Revised verdict on Point 11

Coherent as a discrete response function? Yes, but the analogy to a stat-mech susceptibility is loose. It is a *self-response* (gradient response to its own induced update) divided by an unrelated data-geometry measure; it does not satisfy any FDR identity. The Yaida (2018) and Han et al. (2021, PRE) susceptibilities are the formal analogues and they involve weight-covariance, not feature-density.

Genuinely novel content beyond (batch sharpness $\times$ gradient norm) / (feature-space surprisal)? The instantaneous scalar S_t is **largely redundant**: numerator $\approx$ Bai et al. 2025 $\lambda_{\text{grad}} \cdot \|\mathbf{g}_t\| \approx$ batch sharpness $\times$ gradient norm; denominator is approximately the kernel-residual surprisal in late training and approximately the representation-spectral-entropy of Truong et al. 2026 in mid-training. The novelty is the *coupling*: numerator and denominator *both* depend on θ_t through the current penultimate features, making S_t a coupled dynamical variable rather than a sum of static probes. This coupling has not been studied in the published literature.

Layer-indexed and Fourier-decomposed extensions? The layer-indexed extension is *low novelty* (per-layer Fisher, per-layer sharpness, per-layer Jacobian alignment cover most of the information). The Fourier extension is *medium novelty* — Notsawo 2023 and Hennick-Corlauer 2026 already do Fourier/spectral-heat-capacity diagnostics on training time series, but on different observables; doing Fourier on the specific S_t time series is unpublished.

Worth pursuing as stated, with modifications, or obsolete? Pursue with **substantial modifications**. The strong form of the proposal (define S_t , claim novelty, do a Fourier analysis) would face severe novelty objections from referees who know the Bai 2025, Andreyev-Beneventano 2024, and Notsawo 2023 lines. The weak form (the numerator is a known curvature $\times$ gradient norm; the denominator is a feature-coverage proxy; their joint time series might offer phase-transition forecasting) is defensible.

Strongest formulation that survives cross-examination. Re-cast the project as:

"Cross-spectral diagnostic of training phase transitions: We hypothesise that the **cross-power spectrum** $S_{\{xy\}}(\omega)$ between (a) the batch-directional curvature time series $\lambda_{\text{grad}}(\theta_t) \cdot \|g_t\|$ (Bai et al. 2025; Andreyev & Beneventano 2024) and (b) a feature-space coverage entropy time series $H_B^{(c)}(\theta_t)$ estimated via penultimate-layer KDE (Pichler et al. KNIFE, 2022) carries early-warning information about (i) edge-of-stability onset, (ii) grokking phase transitions, and (iii) loss-of-plasticity in continual learning that neither time series alone provides. We test this against the Notsawo spectral-signature baseline, the Hennick-Corlourer 2RDM baseline, and the Truong-et-al. spectral-entropy baseline." [arxiv](#)

Framed this way, the project is empirically tractable, has clear baselines that would *not* be expected to capture cross-spectral phase-locking between curvature and feature-coverage, and asks a falsifiable question.

Recommendations

Stage 1 — De-risk the numerator (1-2 weeks). Verify empirically that $\|\Delta g_t\|/\eta$ on a single batch correlates $\rho > 0.9$ with batch sharpness $\times \|g_t\|$ (and with the Bai et al. $\lambda_{\text{grad}} \cdot \|g_t\|$ predictor). If correlation is high, then *do not claim* the numerator as new — cite Bai 2025 and Andreyev-Beneventano 2024 and present S_t as a composite. **Threshold to stop: if $\rho > 0.95$, abandon any attempt to claim numerator novelty.**

Stage 2 — Test denominator independence (2-4 weeks). Compute (i) penultimate-KDE feature entropy $H_B^{(c)}$, (ii) the kernel-residual surprisal $\sigma_{\text{NTK}}(B; \theta_t)$, and (iii) Truong et al.'s normalised spectral entropy of representation covariance, on the same training runs. Compute pairwise Pearson and rank correlations through training. **Threshold to continue: if $H_B^{(c)}$ shows $|\rho| < 0.7$ with both (ii) and (iii) for at least one phase of training, the denominator is genuinely distinct enough to justify further work.** If all three are essentially the same observable, replace $H_B^{(c)}$ with the cheaper representation-spectral-entropy of Truong et al. and drop the KDE machinery.

Stage 3 — Run the cross-spectrum test (4-8 weeks). On grokking benchmarks (modular addition, with hyper-parameters from Notsawo et al. 2023 and Truong et al. 2026), pre-training instability runs, and continual-learning plasticity-loss benchmarks (Dohare et al. Nature 2024 ImageNet-CL setup), compute the cross-power spectrum and coherence of (numerator) and (denominator) time series. Compare prediction lead times against: [Nature](#)

- Notsawo spectral signature baseline (single-curve Fourier);
- Hennick-Corlourer spectral heat capacity (single-curve, 2RDM);
- Truong spectral-entropy threshold crossing.

Threshold to declare success: the cross-spectrum lead time on grokking onset must exceed the better of {Notsawo, Truong} lead times by a margin larger than seed variance, on ≥ 3 distinct task families.

Stage 4 — Drop or reframe the layer-indexed extension. Per-layer $S_t^{(\ell)}$ almost certainly recapitulates per-layer Fisher (Achille) and per-layer Jacobian alignment (Lowell-Kastner). Only pursue if Stage 3 succeeds and the cross-spectral signal is *localised* to specific layer indices.

What would change these recommendations:

- If a peer-reviewed paper appears defining exactly $S_t = \|\Delta g_t\| / H_B^{(c)}$ before this project ships, abandon as obsolete.
- If $H_B^{(c)}$ turns out to be $\rho > 0.95$ correlated with Truong et al.'s spectral entropy, drop the KDE component and use the cheaper estimator.
- If the cross-spectrum experiment in Stage 3 fails to beat single-curve baselines on any task family, the research direction is dead.

Caveats

- **Citation hygiene.** Bai et al. 2506.04805, Andreyev-Beneventano 2412.20553, Notsawo et al. 2306.13253, Hennick-Corlour 2603.29805, Truong et al. 2604.13123, Xu 2603.28964/2603.15678, Lowell-Kastner 2406.00127, Bolshim-Kugaevskikh 2510.17486, GradientStabilizer 2502.17055, Lin 2510.00468 are **all preprints**, not peer-reviewed. Several are very recent (2026) and have not been independently replicated. The peer-reviewed anchors are: Damian-Nichani-Lee (ICLR 2023), Cohen et al. (ICLR 2021), Atanasov-Bordelon-Pehlevan (ICLR 2022), Kumar-Bordelon-Gershman-Pehlevan (ICLR 2024), Doan et al. (AISTATS 2021), Achille-Rovere-Soatto (ICLR 2019), Bordelon-Canatar-Pehlevan (ICML 2020), Rosenfeld-Risteski (ICLR 2024), Yaida (ICLR 2019), Hiratani (NeurIPS 2024), Dohare et al. (Nature 2024), Han-Park-Lee-Han (Phys. Rev. E 2021), Pichler et al. (ICML 2022 / KNIFE), Paul et al. (NeurIPS 2021 / GraNd-EL2N), Mindermann et al. (ICML 2022 / RHO-LOSS). [Proceedings of Machine Le...](#)
- **The Bai et al. 2025 preprint is the most consequential single citation for this analysis.** If it is later retracted or its central claim about λ_{grad} 's role in Adam loss spikes is overturned, the case that $\|\Delta g_t\|$ is "essentially already published" weakens.
- **The denominator's coupling claim ("makes S_t a coupled dynamical quantity") is plausible but not demonstrated.** The actual empirical *separability* of numerator and denominator dynamics is an open question. Stage 2 of the recommendations is the test.
- **Statistical-mechanics rhetoric is dangerous in ML papers.** Reviewers will probe whether S_t satisfies any FDR-like identity; it does not. Frame the work as "discrete response function" and avoid claims that S_t is a stat-mech susceptibility. The Lindner (2204.00549) and Yaida (1810.00004) papers are the appropriate models for how to invoke FDR-style language responsibly.
- **The Fourier/spectral-diagnostic field is moving fast in 2025–2026.** A 6-month delay risks being scooped on the cross-spectrum idea by one of the active groups (Pehlevan-Bordelon, Iliad, Beneventano-Poggio, Barkeshli, the 2RDM authors). Stage 1–3 should be compressed if pursued.